Department of Mathematics, IIT Madras

Scientific Computing Advanced Programming Lab (MA-5105)

QUIZ-IV

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- This question paper has one question and sub-questions, totalling to 15 marks and all are to be answered.
- As announced, malpractice can result in disciplinary actions as per the Senate mandate.

Q.1 Automatic Differentiation (AD) is a computational technique for evaluating derivatives by propagating derivative information through elementary mathematical operations. Consider a function $f : \mathbb{R} \to \mathbb{R}$. In AD, each intermediate variable is associated with a pair $(a, \dot{a})$, where a is its numerical value and $\dot{a}$ is its derivative with respect to the independent variable. For example, if $z = xy$, then the value and derivative of z are computed simultaneously:

$$z = xy, \qquad \dot{z} = x\dot{y} + y\dot{x}$$

. The propagation rules are:

Arithmetic operations:

$$\begin{aligned}(a, \dot{a}) + (b, \dot{b}) &= (a + b, \dot{a} + \dot{b}) \\ (a, \dot{a}) - (b, \dot{b}) &= (a - b, \dot{a} - \dot{b}) \\ (a, \dot{a})(b, \dot{b}) &= (ab, a\dot{b} + b\dot{a}) \\ (a, \dot{a})/(b, \dot{b}) &= (a/b, (b\dot{a} - a\dot{b})/b^2)\end{aligned}$$

Elementary functions:

For a differentiable function g,

$$g(a, \dot{a}) = (g(a), g'(a)\dot{a})$$

In particular,

$$\exp(a, \dot{a}) = (e^a, e^a\dot{a})$$

For a function of several independent variables, the derivative component can be extended to a vector. For example, for three independent variables, initialize

$$\begin{aligned}x &\longrightarrow (x, (1, 0, 0)), \\ y &\longrightarrow (y, (0, 1, 0)), \\ z &\longrightarrow (z, (0, 0, 1)).\end{aligned}$$

The propagation rules remain the same, with scalar derivative quantities replaced by vectors. Consequently, a single evaluation of a function can return its value and all its first-order partial derivatives. Consider the non-linear function of three variables

$$F(x, y, z) = e^{-(x^2+2y^2+z^2)} + \sin(xy) + \cos(yz) + \log(1 + x^2 + y^2 + z^2).$$

Your task is to design an automatic differentiation framework capable of evaluating the gradient of such functions.

(a) **Algorithm and data structure design (5 marks)**

Propose an object-oriented design for representing variables and their associated derivative information. Describe:

- What information should be stored for each variable?
- How should arithmetic operations between variables be handled?
- How should operations involving elementary mathematical functions be incorporated into your framework?

Present your proposed design using suitable pseudocode. Your design should be general enough to handle functions of several independent variables.

(b) **Operator implementation (4 marks)**

Write pseudocode for the implementation of the following operations within your proposed framework:

- Multiplication of two variables.
- Division of two variables.
- Evaluation of the exponential function.
- Evaluation of the sine function.

Your pseudocode should show how both the function value and derivative information are updated.

(c) **Application to gradient computation (4 marks)**

Using your proposed framework, describe an algorithm to evaluate:

$$\nabla F(1, 1, 1)$$

- Your algorithm should compute all three partial derivatives in a single evaluation of the function.
- Clearly specify how the independent variables are initialized and how the final gradient is extracted.

(d) Suppose you want to extend your framework to evaluate the Hessian matrix of F. Briefly discuss what additional derivative information or modifications would be required. **(2 marks)**